

3D Printed Footshell for Improved Prosthetic Functionality

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Engineering a Modular, Additively Manufactured Footshell

I. PROBLEM

Prosthetic foot shells protect the keel (see figure 2) and provide the function and appearance of a natural foot, but current designs create major usability and durability challenges.

Current Problems with Foot Shells

- Installation takes significant effort and manipulation, which can lead to material damage
- Traction is limited on smooth or wet surfaces, reducing overall stability
- Water can collect inside the cavity, affecting safety and user independence
- Materials wear down quickly, leading to more frequent replacements



Figure 1. Conventional foot shell Figure 2. Prosthetic keel

II. SPECIFICATIONS

Functional Requirements

- Multi-part design for easier installation
- Treaded bottom for slip resistance
- Split-toe for sandal compatibility and barefoot control
- Water-resistant with integrated drainage

Performance Requirements

- Supports body weight for an average size male patient
- Allows natural gait
- Maintains integrity under repeated loading
- Compatible with multiple prosthetic models

Manufacturing Requirements

- Fully additively manufactured
- Scalable across sizes
- Material must balance strength, flexibility, and cost

III. DESIGN FEATURES

Modular Multi-Part Footshell

- Front and rear sections close around the prosthetic keel
- Enables fast, low-effort, safer installation
- Individual components can be replaced to extend foot shell lifespan

Treaded Underside for Traction

- Textured sole improves grip on wet and smooth surfaces
- Maintains flexibility for natural gait and barefoot maneuverability

Split-Toe Geometry

- Supports sandal/flip-flop compatibility
- Enhances barefoot control and stability

Water-Resistant TPU & Drainage Geometry

- MJF-printed TPU reduces water retention
- Internal geometry allows drainage
- Improves comfort during showering, swimming, and daily use

Iterative FDM Prototyping

- Rapid prototyping refined strength, flexibility, and manufacturability
- Ensured accessibility for clinicians and end-users

User-Centered Improvements

- Designed for faster, safer installation in clinical settings
- Geometry supports natural movement and reduces user effort



Figure 3. Isometric view of design Figure 4. Treaded sole of design Figure 5. Exploded view of design

IV. TESTING

Performed client testing on 2 iterative FDM Prints. First tested foot shell flexibility, support, strength, and client gait. Produced an additional iteration to adjust prosthetic to client feedback and test the foot shell's strength, support, and slip on a 12° incline.



Figure 6. FDM printed footshell in material flexibility test. Figure 7. FDM printed footshell fully assembled in client testing.

Figure 8. FDM printed footshell supporting full client weight in 12° incline testing

V. RESULT ANALYSIS

Fit and Interface Performance

- Overall fit was stable; only minor adjustment needed was the slightly undersized anterior heel cavity.
- Additional clearance around prosthetic set screws is recommended to improve donning and reduce long-term wear.

Mechanical Behavior During Gait

- Footshell demonstrated appropriate elastic response, flexing and fully returning to shape during normal walking.

Durability Under Load

- Survived repeated 10 ft walking cycles and concrete loading with only minor damage.
- All failure was confined to a known FDM support-removal stress point, which will be eliminated in the MJF version and did not impact performance.



Figure 9. Final MJF Printed footshell Figure 10. Final FDM Printed footshell

VI. CONCLUSION & RECOMMENDATIONS

Design Adjustments for MJF Transition

Adjusted rail-system tolerances to accommodate higher surface friction in MJF compared to FDM.

Weight Reduction and Flexibility

Hollowed heel and toe regions and added an internal drain path to remove unfused MJF powder and reduce mass.

Feature Reinforcement

Strengthened drainage-hole walls to address increased cracking risk due to lower elasticity in MJF material.

Structural Optimization

Explored lattice integration in hollow regions or shell surfaces to reduce weight while maintaining structural integrity.